

Grupo Promesa

Forecasting recycling demand to
support optimized monthly
collection planning

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Agenda

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- 02 Project Scope
- 03 Work Steps / Analysis
- 04 Key Deliverables
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01 Company Overview

PROMESA is a recycling solutions company that partners with schools, businesses, and cafés to collect and manage recyclable materials responsibly.

Our mission is to drive environmental impact through efficient collection, responsible processing, and community engagement.



Sustainable
Impact



Diverse
Partners



Efficient
Operations



Data-Driven
Decisions



Recycling Network

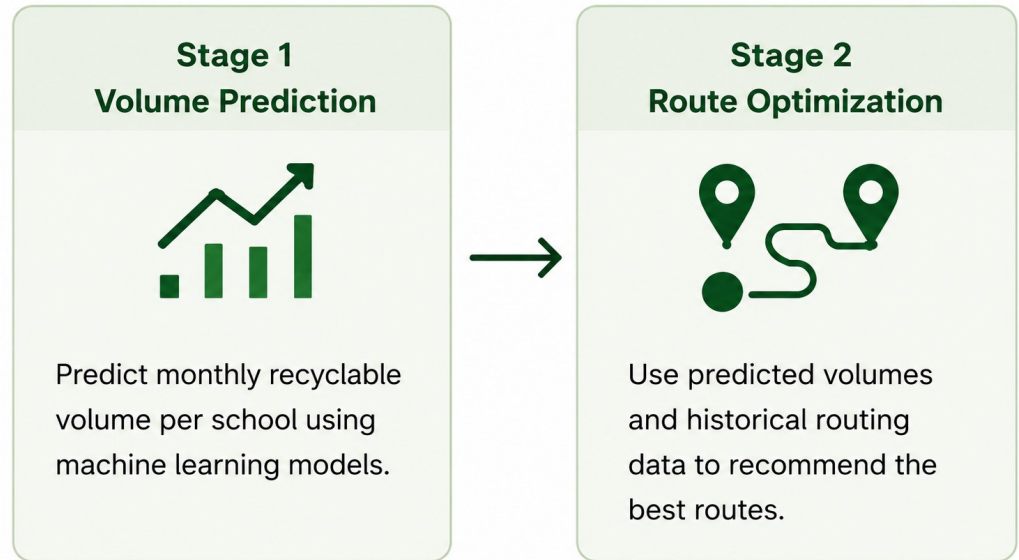
Promesa collects recyclables from multiple partner types across the city through optimized routes.



02 Project Scope

Goal: A two-stage data-driven approach

Develop a data-driven solution to **forecast** the **amount** of recyclable materials collected from schools and provide the foundation for **route optimization**.



Note: Stage 2 was not developed due to data unavailability.

03 Work Steps / Analysis

Stage 1: Data Understanding



Data Collection

Gathered recycling data



Data Cleaning

Handled missing values, outliers, and inconsistencies



Exploratory Analysis

Analyzed trends, seasonality, and key patterns



Feature Engineering

Created time-window features for model training

School Dataset Overview



132

Total Schools

schools in total



114

Usable Schools

**with minimum
required records**



15 Months

Time Span

Aug 2024 - Oct 2025



~9 months

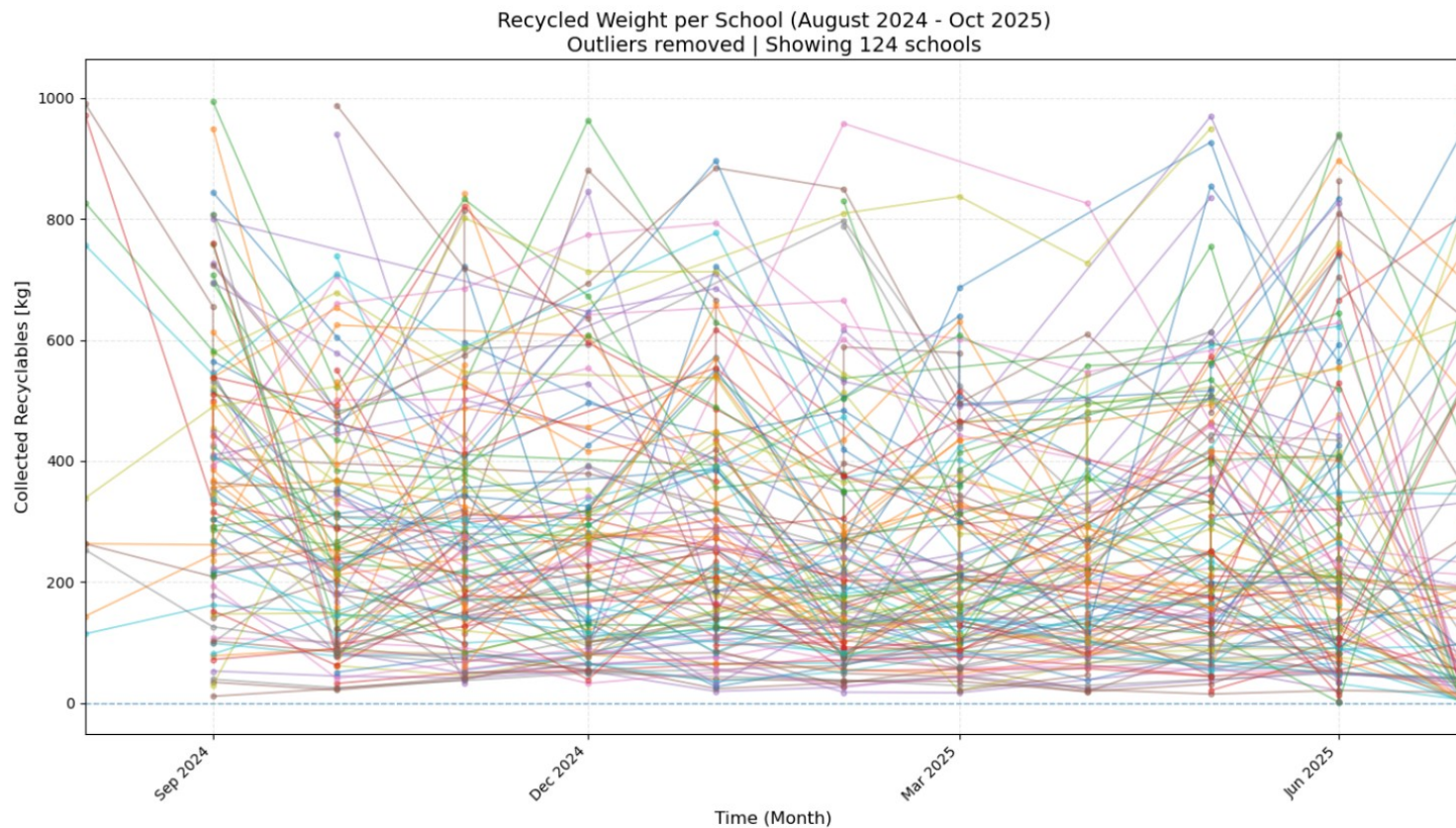
Avg. Data Length

data per school

Key Insight

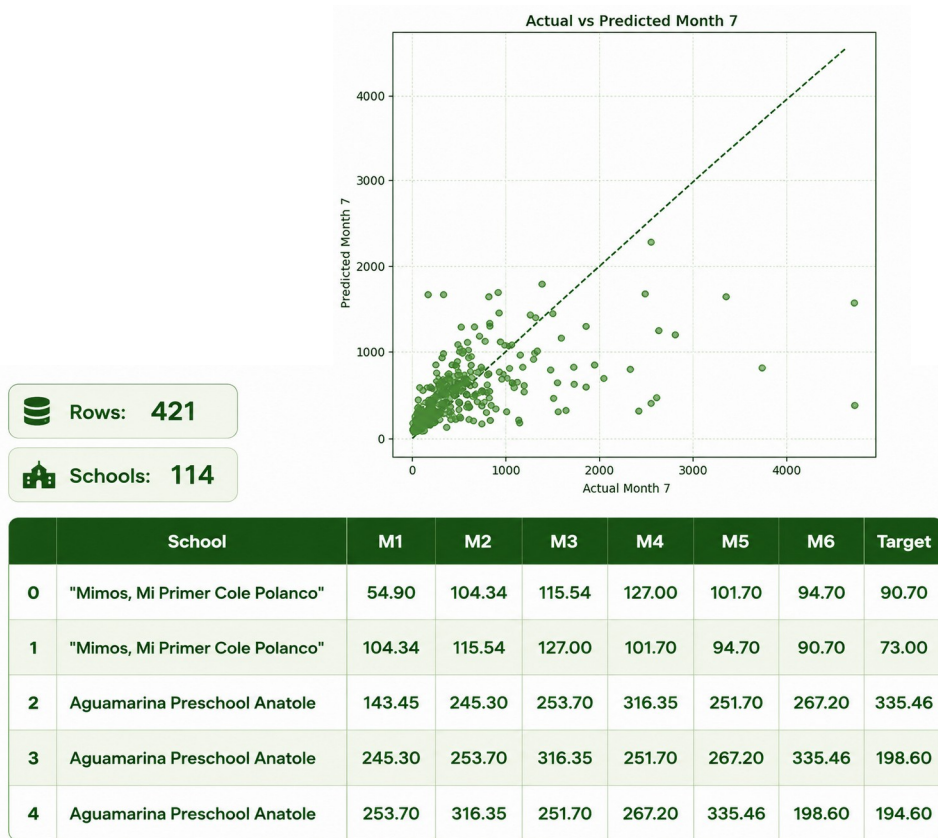
With only 9 months of data, doing long-term forecasting was challenging. So, we focused on short-term monthly prediction for operational planning.

Recycling Collection Over Time (Schools)



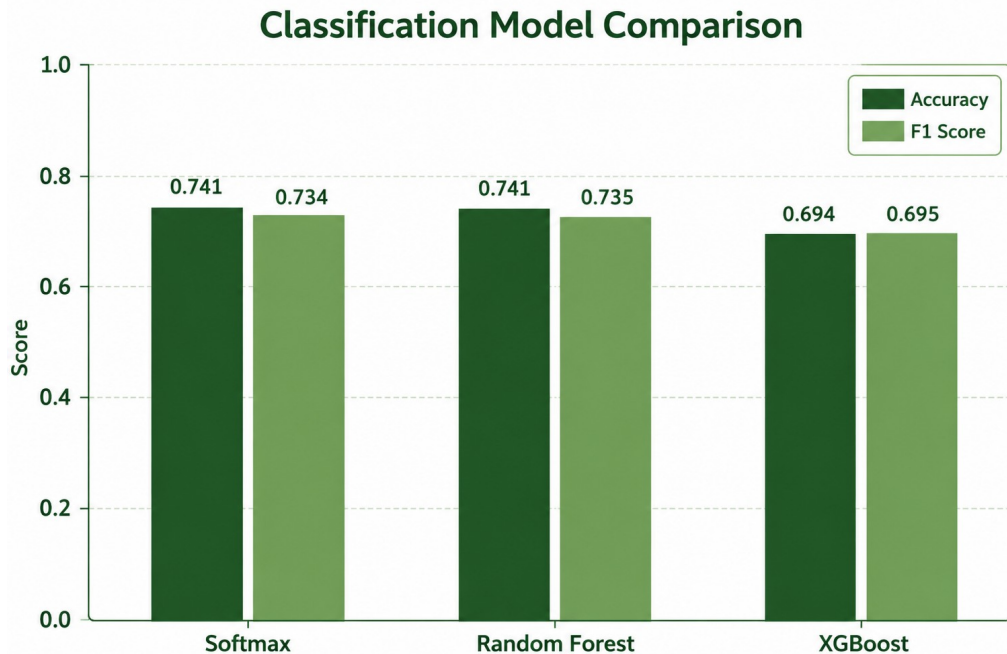
Linear Regression Model

- Created a sliding-window dataset using six consecutive months as predictors and the seventh month as the target.
- Used school recycling totals (kg) from the Promesa school dataset.
- Model achieved $R^2 = 0.334$, meaning earlier months explain about 33.4% of Month 7 variation.
- The model captures general trends but struggles with larger outlier collections.



Discrete Classification Approach

Converted predicted volumes into three classes



Classification performance was strong across setups, providing actionable insights for operational planning.

Café Dataset Overview



61

Cafés
in total



58

Usable Cafés
with minimum
required records



21 Months

Time Span
Jan 2024 – Oct 2025



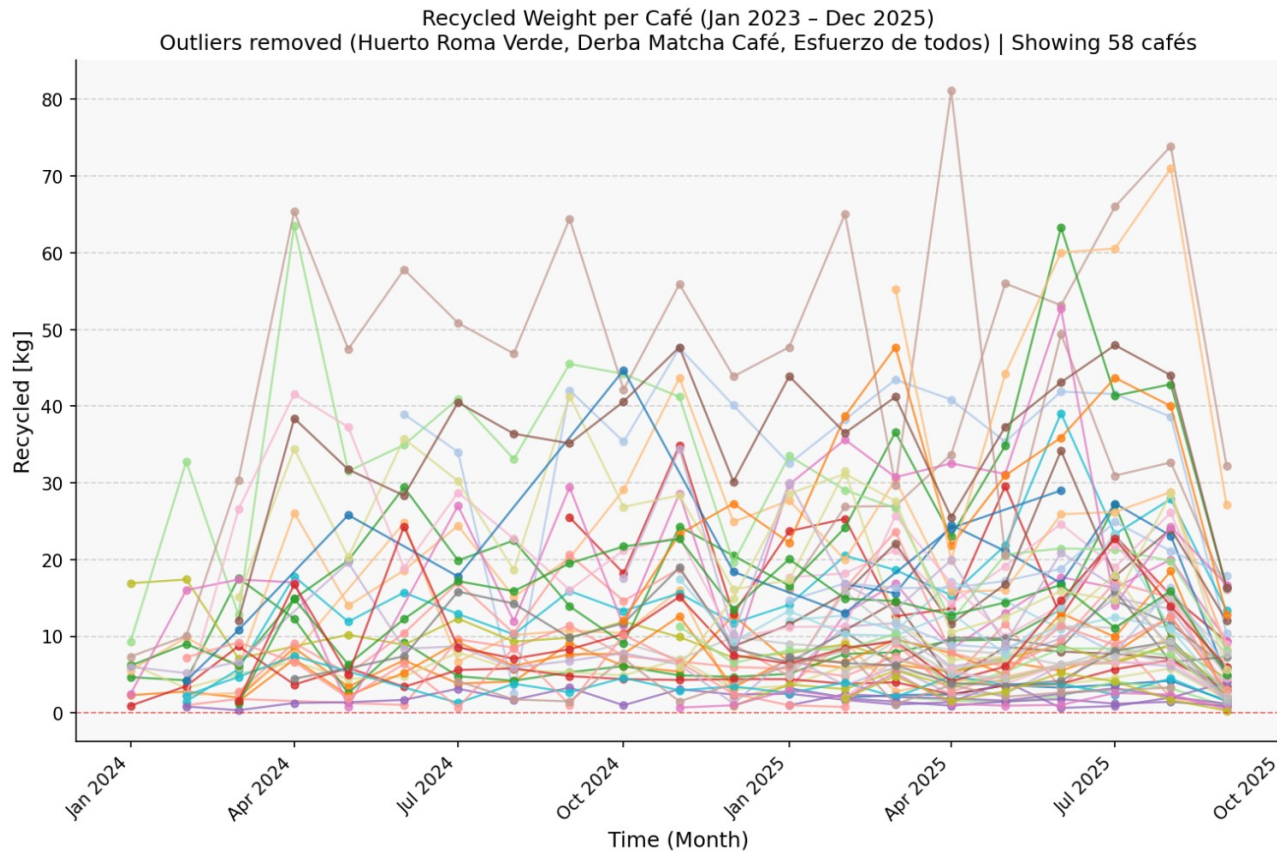
~14 months

Avg. Data Length
data per cafe

Key Insight

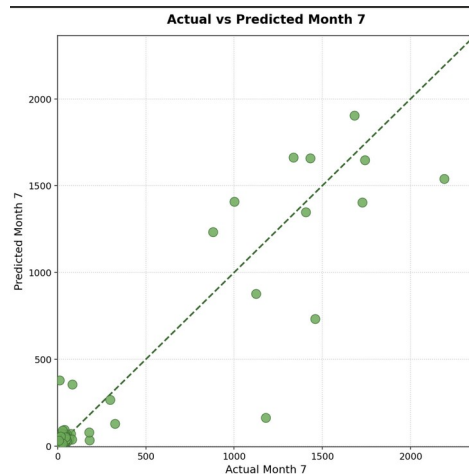
Given the limited temporal depth, uneven reporting frequency, and high inter-café variability, the dataset does not currently support reliable time-series forecasting. The analysis is therefore focused on descriptive and exploratory insights rather than predictive modeling.

Recycling Collection Over Time (Cafés)



Linear Regression Model

- Created a sliding-window dataset using six consecutive months as predictors and the seventh month as the target.
- Used café recycling totals (kg) from the Promesa Café dataset.
- Model achieved $R^2 = 0.8808$, meaning earlier months explain about 88% of Month 7 variation.
- The model captures general trends but struggles with larger outlier collections.



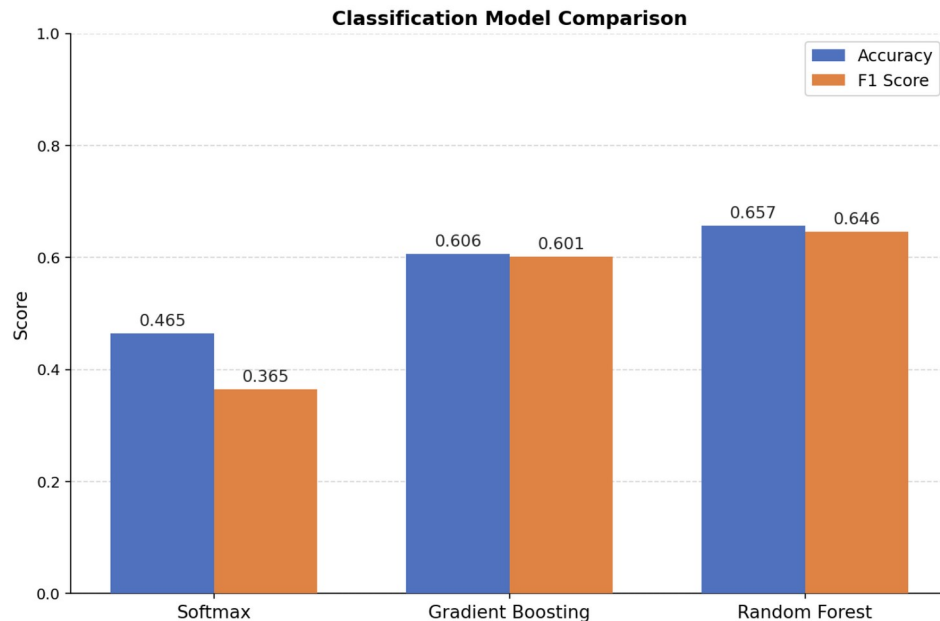
Rows: 491

Cafés: 60

	Café	M1	M2	M3	M4	M5	M6	Target
0	7 CAFÉ	3.2	2.0	2.33	2.4	3.6	3.3	3.7
1	Ahau	38.96	34.02	2.62	42.07	35.36	47.59	40.1
2	Alma Café	2.31	2.79	2.0	8.48	3.42	6.87	4.1
3	Almaquieta	2.34	2.6	7.8	24.78	6.7	10.16	10.74
4	Alquimia Juárez	4.66	4.25	15.08	12.28	8.23	6.0	4.78

Discrete Classification Approach

Converted predicted volumes into three classes



Classification performance was strong across setups, providing actionable insights for operational planning.

Business Dataset Overview



56

Total Businesses

with data collected



33

Max Data Duration

in months



17.6

Avg Data Duration

in months



609.9 kg

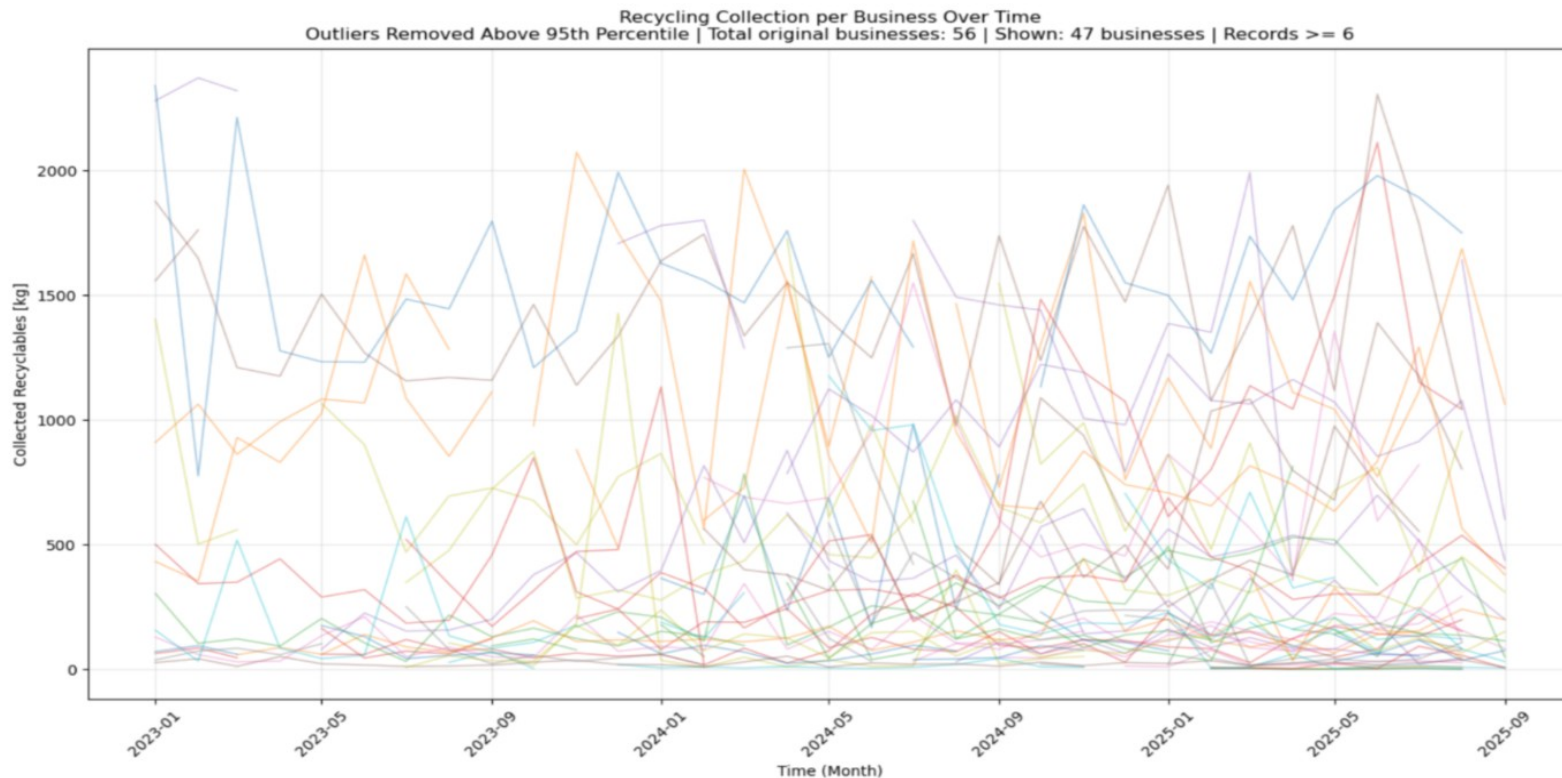
Avg Monthly Value

in kg per business

Key Insight

While the overall monthly average is high (609.9 kg), a small number of businesses contribute disproportionately, with a maximum monthly value of 11820.26 kg. Median-based analysis (238.1 kg) should be used for representative business values.

Recycling collection per business over time



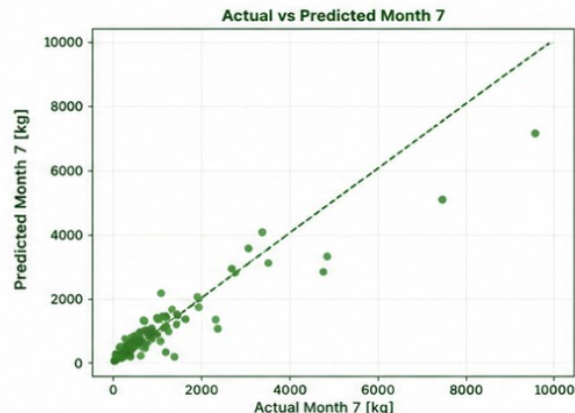
Linear Regression Model

(Business Dataset)

- Created a sliding-window dataset using six consecutive months as predictors and the seventh month as the target.
- Used business recycling totals in kg from the *Empresa PROMESA* dataset.
- Model achieved $R^2 = 0.853$, meaning earlier months explain about 85.3% of Month 7 variation.
- The model captures general trends well but struggles with very high-volume business outliers.

 Rows: 684

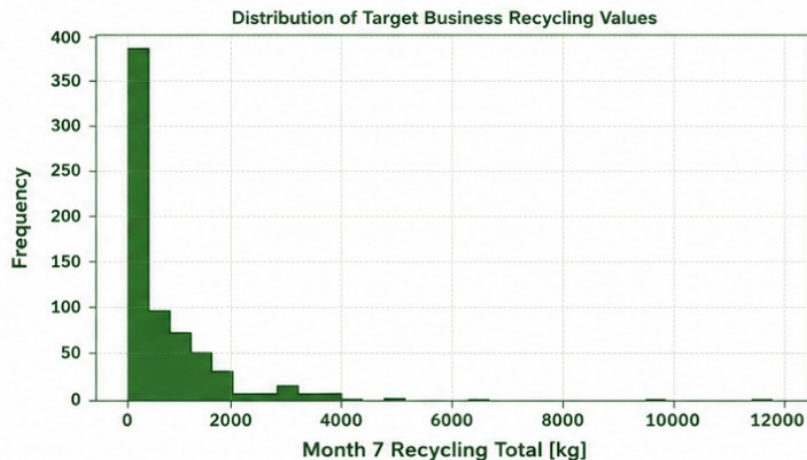
 Businesses: 45



	Business	M1	M2	M3	M4	M5	M6	Target
0	AMEX	364.72	299.96	696.46	236.30	687.78	165.91	984.10
1	AMEX	299.96	696.46	236.30	687.78	165.91	984.10	239.68
2	AMEX	696.46	236.30	687.78	165.91	984.10	239.68	780.22
3	Basalto	908.60	1062.70	861.80	992.40	1083.00	1068.00	1585.50
4	Basalto	1062.70	861.80	992.40	1083.00	1068.00	1585.50	1282.50
5	Basalto	861.80	992.40	1083.00	1068.00	1585.50	1282.50	2492.60
6	Basalto	992.40	1083.00	1068.00	1585.50	1282.50	2492.60	975.60
7	Basalto	1083.00	1068.00	1585.50	1282.50	2492.60	975.60	2073.10
8	Basalto	1068.00	1585.50	1282.50	2492.60	975.60	2073.10	1750.50
9	Basalto	1585.50	1282.50	2492.60	975.60	2073.10	1750.50	1475.10

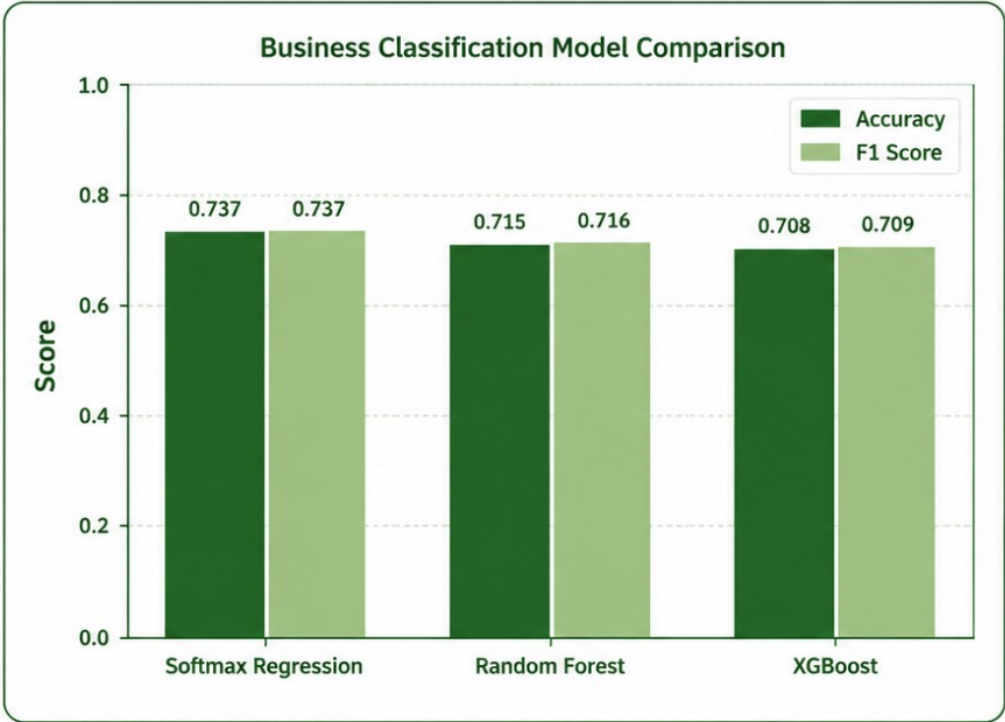
Classification on Business Dataset

- Business recycling values were discretized into tertiles: **Low**, **Medium**, and **High** using `pd.qcut()`.
- The distribution is right-skewed — most businesses recycle at moderate monthly volumes, while a small number of businesses create high-volume outliers.
- Three classification models were compared: **Softmax Regression**, **XGBoost**, and **Random Forest**.
- The winner, **Softmax Regression**, achieved the highest overall performance with an accuracy of around **0.737** and F1 score around **0.737**.
- Earlier monthly recycling patterns provide predictive value, but boundary overlap between Medium and High/Low categories may still affect classification accuracy.



	precision	recall	f1-score	support
High	0.81	0.78	0.80	45.00
Low	0.78	0.83	0.80	46.00
Medium	0.62	0.61	0.62	46.00
accuracy	0.74	0.74	0.74	0.74
macro avg	0.74	0.74	0.74	137.00
weighted avg	0.74	0.74	0.74	137.00

Classification on Business Dataset



Model		Accuracy	F1 Score
0	Softmax Regression	0.737226	0.736519
1	Random Forest	0.715328	0.716333
2	XGBoost	0.708029	0.709487

04 Key Deliverables



Cleaned & Structured School Dataset

9 months of historical
data for 114 schools



ML Model for Volume Prediction

Predicts next month
collection per school



Discrete Volume Classification

Low / Medium / High
volume categories



Insights & Recommendations

Actionable insights to
support operations

05 Project Benefits to Promesa



Better Planning

Accurate volume forecasts for resource planning



Operational Efficiency

Improved scheduling and route planning (when Stage 2 is implemented)



Cost Savings

Optimized resource allocation and reduced costs



Environmental Impact

More recycling collected, greater impact



Data-Driven Decisions

Stronger decisions based on data and analytics

Limitations & Next Steps

Limitations

- Limited historical data (only 9 months).
- Route optimization model (Stage 2) could not be developed due to lack of routing data access.
- External factors (holidays, events, policy changes) not included.

Next Steps

- Collect more historical data to improve forecasting accuracy.
- Obtain routing data to build and deploy optimization model.
- Explore advanced models and external data integration.

06

Thank You!

Questions & Discussion

